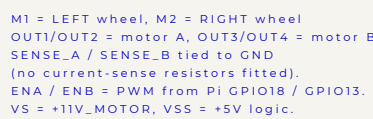


1. POWER / BATTERY / REGULATORS



*** DO NOT CONNECT 11.1V TO RASPBERRY PI ***
U1 5V pins are fed ONLY from U5 +5V.

2. MOTOR DRIVER L298N



ECHO DIVIDER (mandatory level shift)
HC-SR04 ECHO = 5V logic, Pi GPIO = 3.3V
R1 1kΩ / R2 2kΩ -> ECHO_3V3 = 3.33V
(5V x 2k / 3k). NEVER wire ECHO direct.

```
U1 GPIO MAP (all 26 GPIOs used)
I2C GPIO2/SDA + GPIO3/SCLE -> U3 PCA9685
SPI0 GPIO10/9/11 + CE0/CE1 -> J32 TFT
CE0 = TFT_CS, CE1 = touch_T_CS
ENA/ENB = PWM GPIO18 / GPIO13
UART GPIO14/15 <-> U4 ESP32-CAM (crossed)
5V from US ONLY - NEVER 11.1V
BT audio -> Dubstep Pot 600 (wireless)
```

```

U0T (GPIO1) = TX -> Pi RXD = ESP32_TX
U0R (GPIO3) = RX <- Pi TXD = ESP32_RX
Camera / person detection / inference on Pi
5V from +5V rail, GND common, rest NC.

```

U3 PCA9685PW: A0-A5 = GND (addr 0x40),
~OE = GND (outputs enabled), EXTCCLK = NC.
VDD = +5V logic. Servo V+ rail = +5V_SERVO
routed to J4-J7 headers. LED0-3 = CH0-3.

J2 = 2.4in ILI9341 240x320 SPI display with XPT2046 resistive touch. Touch shares the SPI bus, T_CS on CE1 (GPIO7). BL = GPIO20 backlight, VCC = +5V.

SERVO POWER RAIL: +5V_SERVO from U6 only.
Header: 1 SIGNAL . 2 +5V_SERVO . 3 GND
(common ground). NEVER from Pi 5V GPIO.

FLYSKY FS-i6X - WIRELESS TRANSMITTER
(2.4 GHz) binds to J1 FS-iA10B receiver.
CH1-CH4 drive / camera, CH5 = NAMASTE.
Pi reads the PWM channel pulses on GPIO.

Dubstep Pop 600 speaker: audio link is BLUETOOTH (wireless) from the Raspberry Pi. Only +5V / GND are wired (power/charge).

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